

---

# FPGA Experiment Report

## Experiment 5 — PS/2 Keyboard Interface

---

**Task:** FPGA 5 — Keyboard (PS/2 interface)

**Student 1:** Mahmood Stitia

**ID:** 214032682

**Student 2:** Saleh Khalil

**ID:** 213310485

---

### Part 1 — Status update

The table below lists the submitted files and their status. The status is one of *Done* (module and simulation written and passing), *Partially done* (started but not fully working — explained in the notes), or *Not started*.

| File / Step        | Status | Notes / comments                    |
|--------------------|--------|-------------------------------------|
| Ps2_Interface.v    | Done   | PS/2 receiver; testbench passes     |
| Ps2_Interface_tb.v | Done   | Self-checking; all scenarios pass   |
| Ps2_Display.v      | Done   | 7-seg + LED strobe (CDC to 100 MHz) |
| Ps2_Top.v          | Done   | Top level, FPGA pin wiring          |
| task1.xdc          | Done   | Pin + clock constraints             |

### Issues

Any errors present at the time of submission are described here (per module / simulation, with what the error means).

- Ps2\_Interface.v:
- Ps2\_Interface\_tb.v:
- Ps2\_Display.v:
- Ps2\_Top.v:
- task1.xdc:

---

### Part 2 — Simulations and answers

This part contains the simulation of Ps2\_Interface with annotated waveforms. (*Per the PDF, only Ps2\_Interface.v is simulated.*)

## Simulation — Ps2\_Interface

[ paste screenshot here ]

Annotated waveform of *Ps2\_Interface\_tb*: *PS2Clk*, *rstn*, *PS2Data*, *scancode[7:0]*, *keyPressed*. Draw arrows on: the reset test, the first make-code, the typematic repeats, the extended key, and the release / re-press.

**What the waveform shows** (annotate the real screenshot with arrows):

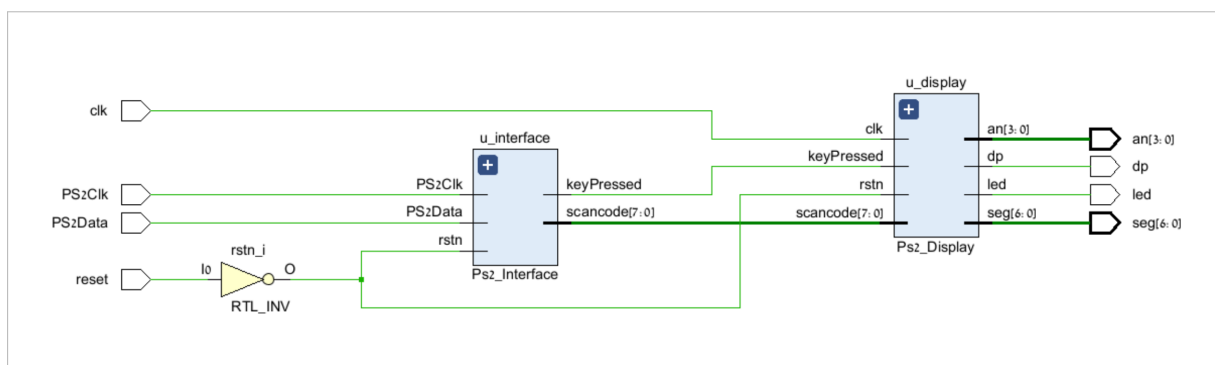
1. **Reset test** — with *rstn*=0 the outputs are cleared even while *PS2Clk* is idle (the async-reset advantage).
2. **First make-code** 0x2E — after the stop-bit edge *scancode* becomes 2E and *keyPressed* pulses exactly once.
3. **Typematic repeats of** 0x2E — the same code is sent twice more; *scancode* stays 2E and *keyPressed* does *not* pulse again.
4. **Extended key** 0xE0, 0x5A — the 0xE0 frame leaves *scancode* unchanged and produces no pulse; the trailing 0x5A sets *scancode*=5A with one pulse.
5. **Release** 0xF0, 0x5A — neither frame pulses; the displayed code is held.
6. **Re-press** 0x5A **after release** — pulses again (a new press after a release is a new event).

[ paste screenshot here ]

Zoom-in showing that the data is sampled on the negative edge of *PS2Clk* (arrow on a falling edge where *scancode* / *keyPressed* react).

## Elaborated Design

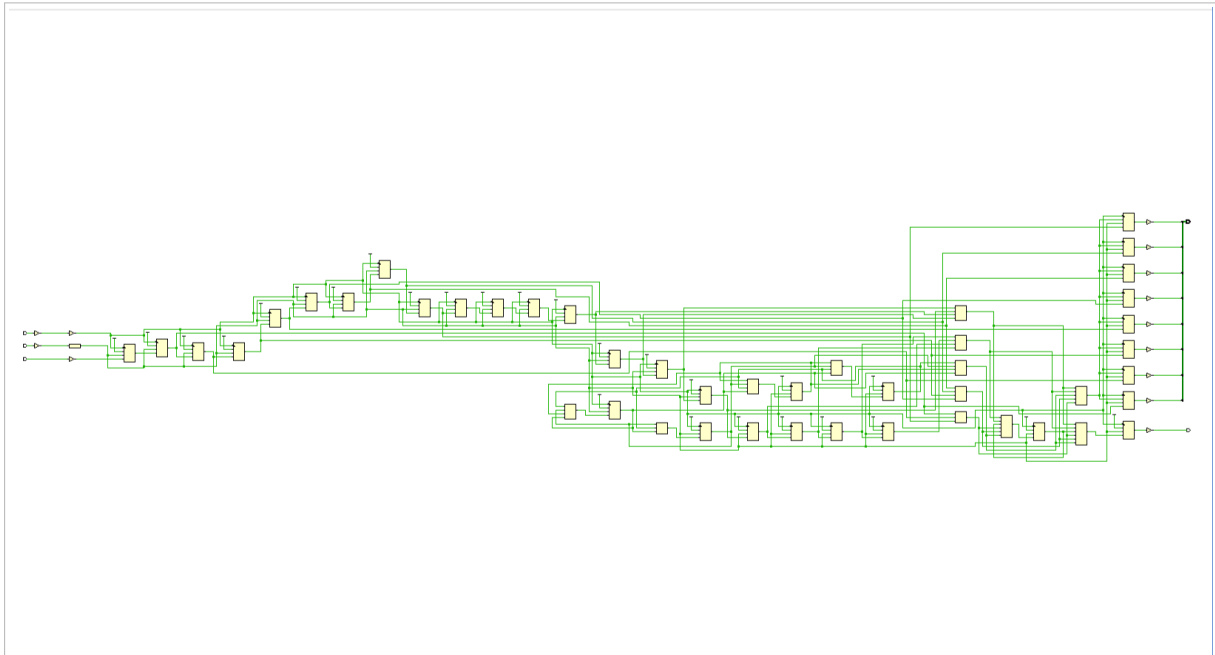
### Top module



RTL-elaborated schematic of *Ps2\_Top*: *u\_interface* (*Ps2\_Interface*, *PS2Clk* domain) feeds *keyPressed* and *scancode[7:0]* into *u\_display* (*Ps2\_Display*, 100 MHz domain); the reset push-button passes through an

*RTL\_INV (rstn\_i) into rstn. Inputs clk, PS2Clk, PS2Data, reset are on the left and outputs an[3:0], dp, led, seg[6:0] on the right.*

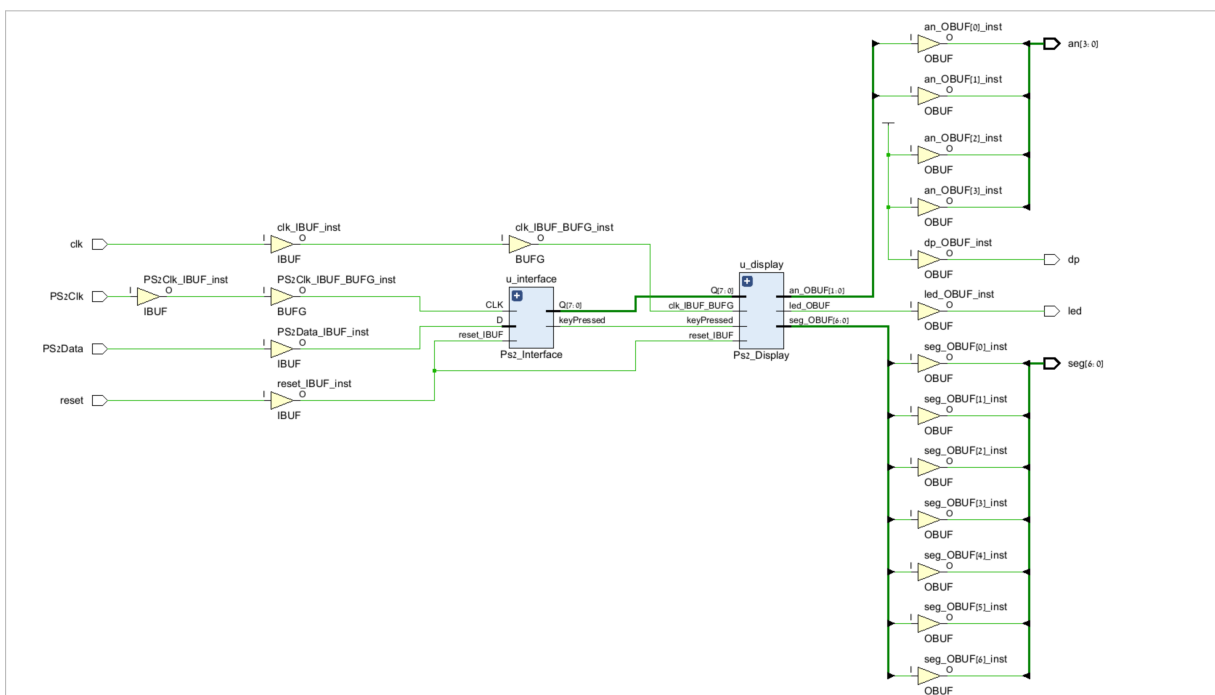
## Ps2\_Interface



*RTL-elaborated schematic of Ps2\_Interface: the 22-bit shift register and the bitcount counter feed the held\_code / key\_down registers and the make / break / extend decode logic, all clocked by PS2Clk; the outputs scanCode[7:0] and keyPressed leave on the right.*

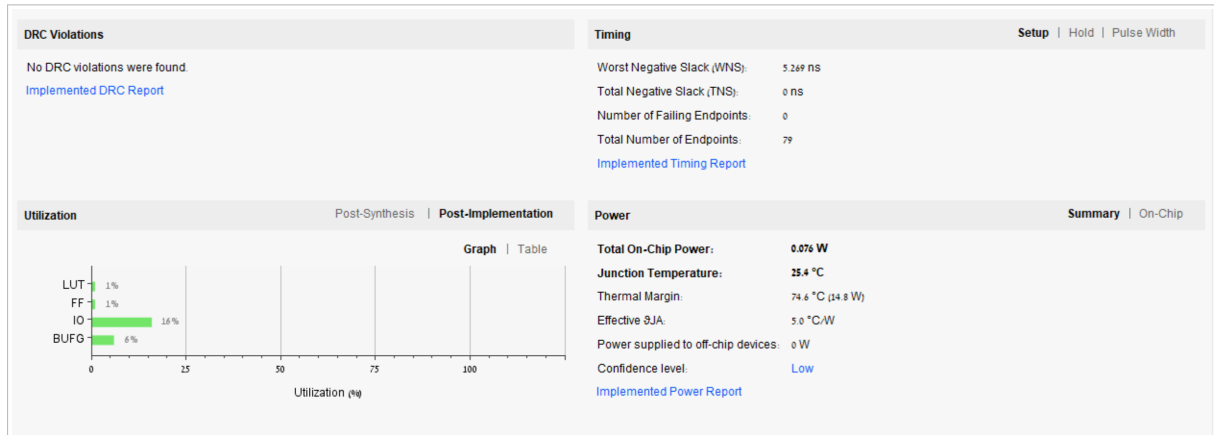
## Implementation

### Synthesis schematic



Post-synthesis schematic. Every external port now passes through an *IBUF/OBUF*, the two clocks (*clk*, *PS2clk*) go through *BUFG* global buffers, and *u\_interface* / *u\_display* drive the *an*, *dp*, *led* and *seg* output buffers.

## Project summary



Post-implementation project summary. No DRC violations; timing met (WNS = 5.269 ns, 0 failing endpoints); utilization is small (LUT & FF  $\approx$  1%, IO 16%, BUFG 6%); total on-chip power is 0.076 W with a junction temperature of 25.4 °C.

## Timing summary

| Design Timing Summary                          |  |                                  |                                                   |
|------------------------------------------------|--|----------------------------------|---------------------------------------------------|
| Setup                                          |  | Hold                             | Pulse Width                                       |
| Worst Negative Slack (WNS): 5.269 ns           |  | Worst Hold Slack (WHS): 0.330 ns | Worst Pulse Width Slack (WPWS): 4.500 ns          |
| Total Negative Slack (TNS): 0.000 ns           |  | Total Hold Slack (THS): 0.000 ns | Total Pulse Width Negative Slack (TPWS): 0.000 ns |
| Number of Failing Endpoints: 0                 |  | Number of Failing Endpoints: 0   | Number of Failing Endpoints: 0                    |
| Total Number of Endpoints: 79                  |  | Total Number of Endpoints: 79    | Total Number of Endpoints: 57                     |
| All user specified timing constraints are met. |  |                                  |                                                   |

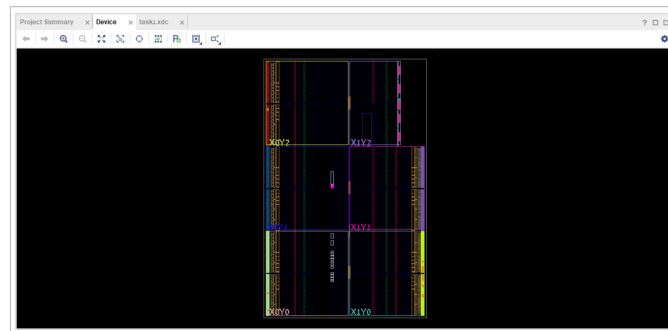
Design Timing Summary after implementation. The Setup (WNS = 5.269 ns), Hold (WHS = 0.330 ns) and Pulse-Width (WPWS = 4.500 ns) slacks are all positive with 0 failing endpoints — “All user specified timing constraints are met”.

## Critical Path

| Name    | Slack | Levels | High Fanout | From                        | To                           | Total Delay | Logic Delay | Net Delay | Requirement | Source Clock | Destination Clock | Exception | Clock |
|---------|-------|--------|-------------|-----------------------------|------------------------------|-------------|-------------|-----------|-------------|--------------|-------------------|-----------|-------|
| Path 1  | 5.269 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[20]CE | 4.361       | 1.054       | 3.307     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 2  | 5.269 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[21]CE | 4.361       | 1.054       | 3.307     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 3  | 5.269 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[22]CE | 4.361       | 1.054       | 3.307     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 4  | 5.269 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[23]CE | 4.361       | 1.054       | 3.307     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 5  | 5.534 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[4]CE  | 4.100       | 1.054       | 3.046     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 6  | 5.534 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[5]CE  | 4.100       | 1.054       | 3.046     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 7  | 5.534 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[6]CE  | 4.100       | 1.054       | 3.046     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 8  | 5.534 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[7]CE  | 4.100       | 1.054       | 3.046     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 9  | 5.576 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[16]CE | 4.087       | 1.054       | 3.033     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |
| Path 10 | 5.576 | 3      | 25          | u_displayled_timer_reg[16]C | u_displayled_timer_reg[17]CE | 4.087       | 1.054       | 3.033     | 10.0        | sys_clk_pin  | sys_clk_pin       |           |       |

Intra-Clock Paths report. The worst paths (Path 1–4) run inside *u\_display* from *led\_timer\_reg[16]C* to *led\_timer\_reg[20..23]CE*, with slack 5.269 ns, 3 logic levels and 4.361 ns total delay (1.054 ns logic + 3.307 ns net) against the 10 ns requirement — comfortably met.

## Device



Device (floorplan) view of the placed-and-routed design on the Artix-7 (XC7A35T). The few used slices are highlighted across the clock regions (X0Y0–X1Y2); the design occupies only a tiny fraction of the fabric.

## Problems we faced

On hardware the design behaves as specified: pressing a numeric-keypad key shows its 8-bit make-code as two hex symbols on the two right-hand 7-segment digits (e.g. 5 → 2E); each new press blinks the LED once with a short strobe (holding the key does *not* re-blink); the displayed code latches until the next press; and extended keys show only their last two hex symbols (the 0xE0 prefix is invisible).

(Describe the problems you actually hit in the lab and how you solved them. For example:)

- (fill in) where to put the 0xF0/0xE0 / two-key handling logic — in the interface vs. the display module — and the wrong display behaviour seen before moving it all into Ps2\_Interface.
- (fill in) the keyboard clock being idle most of the time (decode before the stop bit, async reset).
- (fill in) CDC glitches on keyPressed and debugging the two clock domains in separate ILA cores.

[ paste screenshot here ]

Board photo: keypad plugged into the USB host port, a key pressed and its hex code shown on the two right digits with the LED lit.

## Appendix — Verilog Source Code

The full source of the submitted files is listed below. Paste the code of each file inside its block.

### Ps2\_Interface.v

```
1 // --- paste Ps2_Interface.v here ---
```

**Ps2\_Interface\_tb.v**

```
1 // --- paste Ps2_Interface_tb.v here ---
```

**Ps2\_Display.v**

```
1 // --- paste Ps2_Display.v here ---
```

**Seg\_7\_Display.v**

```
1 // --- paste Seg_7_Display.v here ---
```

**Ps2\_Top.v**

```
1 // --- paste Ps2_Top.v here ---
```

**task1.xdc**

```
1 ## --- paste task1.xdc here ---
```